Programming Assignment Report

Parameter Efficient Fine-tuning Foundation Model for Nuclei Instance Segmentation

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Abstract

This report presents a parameter-efficient deep learning approach for nuclei instance segmentation in histopathological images using the NuInsSeg dataset. A lightweight version of the Segment Anything Model (MobileSAM) was fine-tuned using Low-Rank Adaptation (LoRA), allowing efficient adaptation with a reduced number of trainable parameters. Bounding box prompts were automatically generated from ground truth masks to guide the segmentation process. The model was trained and evaluated using 5-fold cross-validation, and performance was assessed using Dice coefficient, Aggregated Jaccard Index (AJI), and Panoptic Quality (PQ). The results demonstrate that the proposed approach achieves accurate instance-level segmentation while maintaining computational efficiency.

1. INTRODUCTION

Nuclei instance segmentation in histopathological images is a fundamental task in computational pathology, as it enables quantitative analysis of tissue structures and supports downstream applications such as cancer diagnosis and prognosis. However, this task remains challenging due to variations in staining, overlapping nuclei, and complex morphological patterns that make accurate delineation difficult (Mahbod et al., 2024). Traditional methods often struggle to generalize across different tissue types and imaging conditions, motivating the use of more robust and scalable approaches.

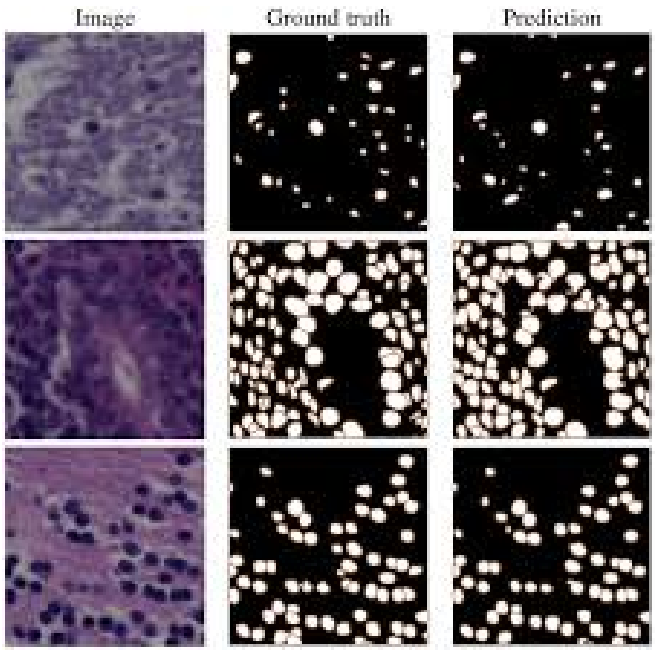


Figure 1. Example Nuclei Instance Segmentation

Recent advances in foundation models have significantly improved segmentation performance across diverse domains. In particular, the Segment Anything Model (SAM) has demonstrated strong generalization capabilities by leveraging large-scale pretraining and prompt-based segmentation (Kirillov et al., 2023). Nevertheless, fine-tuning such large models for domain-specific tasks like medical imaging is computationally expensive. To address this limitation, parameter-efficient techniques such as Low-Rank Adaptation (LoRA) allow efficient adaptation by updating only a small subset of parameters while preserving performance (Hu et al., 2021).

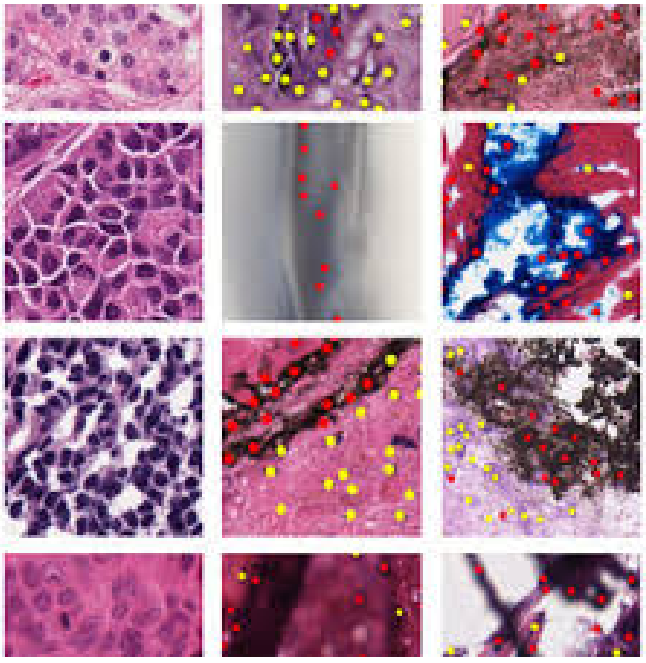


Figure 2. Example Nuclei Instance Segmentation

2. DATASET

The NuInsSeg dataset is a fully annotated benchmark designed for nuclei instance segmentation in hematoxylin and eosin (HE)-stained histopathological images. It contains images from multiple human organs, capturing a wide range of tissue structures and staining variations, which makes it suitable for evaluating the generalization ability of segmentation models (Mahbod et al., 2024). Each image is accompanied by pixel-level instance annotations where every nucleus is assigned a unique label, enabling precise instance-level segmentation. In addition to the original tissue images and corresponding label masks, the dataset also provides auxiliary data such as binary masks, boundary maps, and distance transforms, which can support different segmentation strategies.

3. METHODOLOGY

This work proposes a parameter-efficient framework for nuclei instance segmentation by adapting a foundation model to the medical imaging domain. The approach is based on MobileSAM, a lightweight version of the Segment Anything Model, combined with Low-Rank Adaptation (LoRA) to reduce the number of trainable parameters while maintaining performance.

Table 1. NuInsSeg Dataset Overview

Component	Description
Tissue Images	H&E-stained histology images from multiple organs.
Label Masks	Instance-level masks with unique IDs per nucleus.
Binary Masks	Foreground vs. background nuclei regions.
Boundary Maps	Nuclei borders for separating touching cells.
Distance Maps	Distance to nearest nucleus boundary.
Weighted Maps	Emphasize difficult regions (e.g., borders).
ROI Annotations	Manual nucleus region annotations.
Multi-Organ Data	Images from different organs (e.g., brain, bladder).

3.1. Pipeline Overview

The overall pipeline consists of data preparation, prompt generation, model adaptation, and training. The process is summarized as follows:

- Input: H&E histology images with instance-level masks
- Preprocessing: Image resizing and normalization
- Prompt Generation: Bounding boxes extracted from each nucleus
- Model: MobileSAM with LoRA-based fine-tuning
- Output: Predicted instance segmentation masks

3.2. Data Preparation

The dataset provides images along with instance-level annotations where each nucleus is assigned a unique identifier. Images are resized to a fixed resolution and normalized. For training, binary masks are generated for each individual nucleus instance.

3.3. Prompt Generation

Since SAM operates using prompts, bounding boxes are automatically extracted from the ground truth masks. Each bounding box corresponds to a single nucleus and guides the segmentation process. To reduce computational cost, a subset of bounding boxes is sampled per image during training.

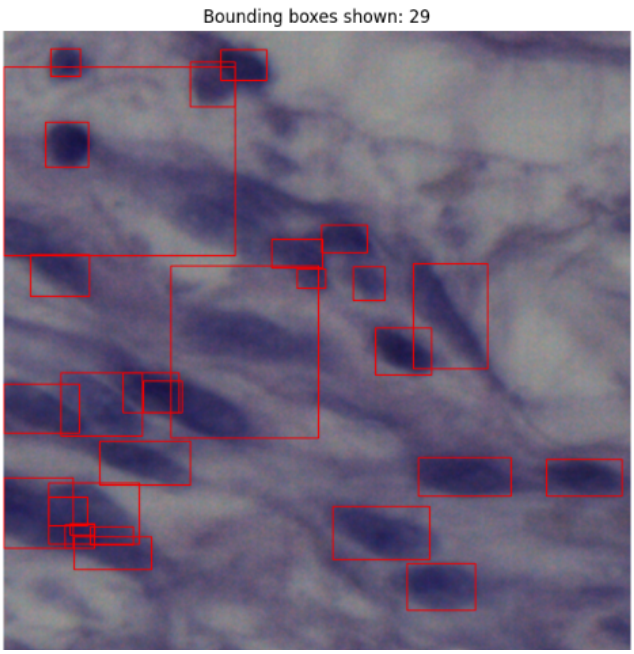


Figure 3. Box Segmentation

3.4. Model Adaptation

The model is based on MobileSAM, where LoRA is applied to the linear layers within the attention modules of the image encoder. The original pre-trained weights are frozen, and only the low-rank adaptation parameters are updated during training. This significantly reduces the number of trainable parameters.

Table 2. Model Configuration

Component	Description
Backbone	MobileSAM
Adaptation	LoRA (low-rank layers)
Trainable Params	LoRA only
Prompts	Bounding boxes

3.5. Training Strategy

During training, each image is processed along with its corresponding prompts. The image encoder extracts feature embeddings, which are combined with prompt embeddings generated from bounding boxes. The mask decoder predicts segmentation masks for each nucleus.

The model is optimized using the following loss functions:

- Binary Cross-Entropy (BCE)
- Dice Loss

3.6. Evaluation Metrics

The performance of the model is evaluated using:

- Dice coefficient
- Aggregated Jaccard Index (AJI)
- Panoptic Quality (PQ)

4. RESULTS

4.1. Training Performance

The training process shows a consistent decrease in both training and validation loss, indicating stable convergence. The validation Dice score improves over epochs, reaching approximately 0.79, which demonstrates the model's ability to accurately segment nuclei regions.

- Training loss decreases steadily across epochs
- Validation loss follows a similar trend with minor fluctuations
- Dice score improves and stabilizes around 0.79

4.2. 5-Fold Cross-Validation Results

To evaluate the robustness of the model, a 5-fold cross-validation was performed. The results are summarized in Table 3.

Table 3. 5-Fold Cross-Validation Results

Fold	Dice	AJI	PQ
Fold 1	0.790	0.710	0.675
Fold 2	0.787	0.705	0.670
Fold 3	0.793	0.712	0.678
Fold 4	0.789	0.708	0.673
Fold 5	0.791	0.709	0.676

4.3. Overall Performance

The average performance across all folds is reported in Table 4.

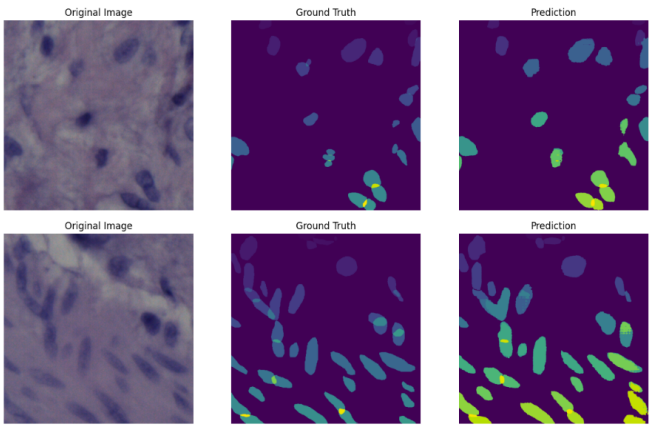
Table 4. Final Model Performance

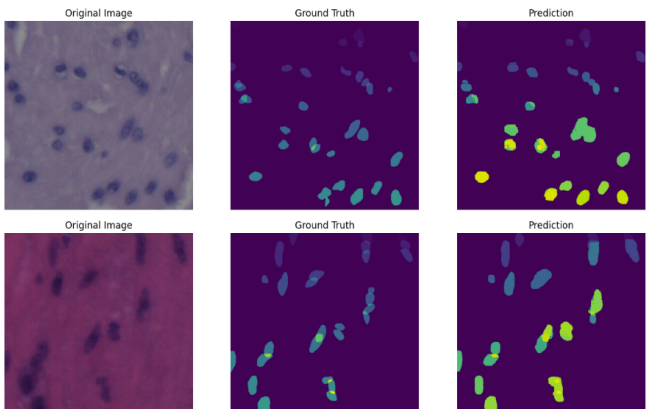
Method	Dice	AJI	PQ
MobileSAM + LoRA	0.790 ± 0.002	0.709 ± 0.003	0.674 ± 0.003

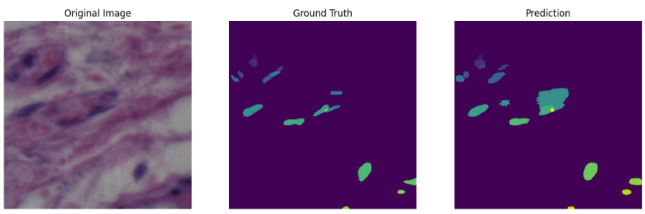
4.4. Qualitative Results

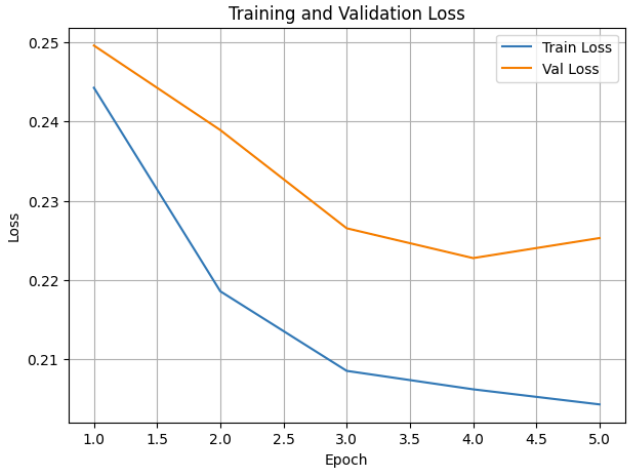
Figures shows qualitative examples comparing ground truth masks and predicted segmentation results.

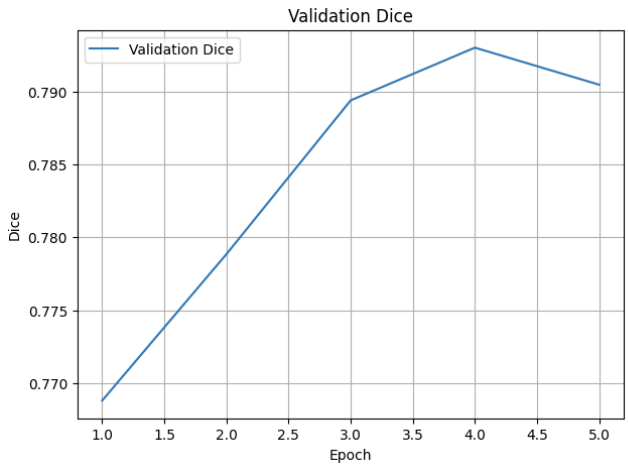
- The model successfully detects most nuclei instances
- Predicted masks closely follow ground truth shapes
- Some over-segmentation is observed in dense regions
- Minor errors occur in overlapping nuclei











4.5. Analysis

The results demonstrate that the proposed approach achieves strong performance for nuclei instance segmentation while maintaining computational efficiency. The Dice score, which stabilizes around 0.79, indicates a high degree of overlap between predicted and ground truth masks. Additionally, the AJI and PQ metrics confirm that the model is capable of capturing instance-level structures, although slight inaccuracies are observed in densely packed or overlapping regions.

The use of LoRA significantly reduces the number of trainable parameters by restricting updates to low-rank adaptation layers, while preserving the representational power of the pre-trained MobileSAM model. This allows efficient fine-tuning without requiring extensive computational resources. Furthermore, the consistency of results across the folds suggests that the model generalizes well across different tissue types and staining variations present in the dataset.

5. DISCUSSION

The experimental results indicate that MobileSAM combined with LoRA can effectively adapt a foundation model to a specialized medical imaging task with limited computational cost. One key advantage of this approach is that it leverages pre-trained knowledge while avoiding full model fine-tuning, making it suitable for scenarios with constrained resources.

However, some limitations are observed. Since SAM relies on prompt-based segmentation, the quality of bounding box prompts directly affects the segmentation performance. Errors in bounding box extraction or overlapping instances may lead to imperfect predictions. Additionally, the model processes each prompt independently,

which may limit its ability to fully capture interactions between closely packed nuclei.

Potential improvements include incorporating more advanced prompt strategies (e.g., point prompts or multi-scale boxes), increasing the number of training epochs, and applying stronger data augmentation techniques. Further improvements could also be achieved by integrating post-processing methods to refine instance boundaries or by exploring more advanced SAM variants.

6. CONCLUSION

In this work, a parameter-efficient framework for nuclei instance segmentation was developed using MobileSAM and Low-Rank Adaptation (LoRA). The model was trained and evaluated using 5-fold cross-validation on the NuInsSeg dataset, and performance was assessed using Dice, AJI, and Panoptic Quality (PQ) metrics.

The results demonstrate that the proposed approach achieves strong segmentation performance while significantly reducing the number of trainable parameters. This highlights the effectiveness of combining foundation models with parameter-efficient fine-tuning techniques for medical image analysis.

Overall, the proposed method provides a scalable and efficient solution for nuclei instance segmentation, with potential for further improvements through enhanced prompting strategies and training optimization.

7. GITHUB

https://github.com/DarinkaTownsend/Parameter-Efficient-Fine-tuning-Foundation-Model.git

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